Parikshit Pareek

Ph.D. Student, Smart Grid Optimization

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I am currently a Ph.D. student and Research Associate working on smart grid optimization and stability assessment methods. My goal is to remove the computational hurdles present in adaptation of green energy sources for electricity. I wish to find the solutions which will enable us to provide safe, affordable and green energy to the person who is at the bottom of the pyramid.

Education

2018-Present **Doctor of Philosophy**, *Nanyang Technological University*, Singapore.
A Novel Framework for Small-Signal Stability Constrained OPF

2016-2018 **Master's of Technology**, *Indian Institute of Technology Delhi*, India.
Computationally Efficient Linear Optimal Power Flow for Power System Applications

2011-2015 **Bachelor's of Technology**, *Rajasthan Technical University Kota*, India.
Electrical Engineering

Awards and Honours

Dr. Shankar Dayal Sharma (Former President of India) Gold Medal-2018, *IIT Delhi.*
Awarded for General Proficiency including Character & Conduct, Excellence in Academic Performance, Extra-Curricular Activities and Social Service among all M.Tech graduating students of IIT Delhi.

POSOCO Power System Award (PPSA)-2019, *POSOCO, India.*
Awarded for Master's Thesis on Power System by Power System Operation Corporation Limited (POSOCO), A Government of India Enterprise.

Prof. O.P Gupta Medal-2018, *IIT Delhi.*
Awarded for being the department rank one at Centre for Energy Studies, IIT Delhi.

Bhagirathi - Bashisht Tiwari Award-2018, *CES, IIT Delhi.*
Awarded for the highest CGPA among all graduating students of M.Tech Energy Studies IIT Delhi.

Shrimati Jawala Devi - Sita Ram Kaushik Award-2018, *CES, IIT Delhi.*
Awarded for best M.Tech Dissertation for the year 2018 at CES, IIT Delhi.

NTU Research Scholarship, *NTU Singapore.*
Awarded for outstanding graduate students for research leading to a Ph.D. at the University.

Research Experience

2018-Present **Graduate Research Scholar**, *Clean Energy Research Lab, NTU Singapore.*
Research Title: A Novel Framework for Small-Signal Stability Constrained OPF.
- Development of sufficient condition for small-signal stability as Bilinear Matrix Inequality.
- Formulation of eigenvalue analysis independent SSSC-OPF based on convexification approach.
- Attacking the SSSC-OPF problem from sharp angles and providing new insights.

2016-2018 **Research Assistant**, *Electrical Energy Systems Lab, IIT Delhi, India.*
Research Title: Computationally Efficient Linear Optimal Power Flow for Power System Applications.
- Development of irregular polygon-based piecewise linearization of quadratic thermal limits.
- Development of renewable forecast based pricing and linear DR-OPF for distribution system.
- Post-Optimal Analysis based method for Solar PV and load uncertainty handling.

2013–2015 **Undergraduate Research Assistant**, *RTU Kota, India.*
Research Title: Condition Monitoring of Squirrel Cage Induction Motor using Stator Current Signal Monitoring

Publications

Parikshit Pareek and Ashu Verma, "Piecewise Linearization of Quadratic Branch Flow Limits by Irregular Polygon" *IEEE Transactions on Power Systems*, Vol. 33, No. 6, pp. 7301-7304, 2018

Parikshit Pareek, Konstantin Turitsyn, Krishnamurthy Dvijotham, Hung D Nguyen, "A Sufficient Condition for Small-Signal Stability and Construction of Robust Stability Region" *IEEE PES General Meeting, August-2019 (In Press)* ArXiv preprint arXiv:1811.03805.

Parikshit Pareek and Ashu Verma, "Computationally Efficient Day-Ahead OPF using Post-Optimal Analysis with Renewable and Load Uncertainties" ArXiv preprint arXiv:1804.06391, 2018

Parikshit Pareek and Ashu Verma, "Linear OPF with Linearization of Quadratic Branch Flow Limits" *IEEE IEEMA Engineer Infinite Conference*, Grater Noida, 2018.

Parikshit Pareek, Nishant Khandelwal, Shashi Raj Kapoor, "A New Approach for Bearing Fault Analysis in Squirrel Cage Induction Motor" *IEEE ICPEICES*-New Delhi, India 2016

Shashi Raj Kapoor, Nishant Khandelwal, Parikshit Pareek , "Bearing Fault Analysis by Signal Energy Calculation based Signal Processing Technique in Squirrel Cage Induction Motor" *IEEE ICSPCT*-India, 2014

Responsibilities

2019-2020 **Director** Outreach Internal Support Committee, EEE Graduate Student Council, NTU Singapore

2018-2019 **Member** EEE Graduate Student Council, NTU Singapore

2016-2018 **Teaching Assistant** Electrical Energy Systems Lab, CES, IIT Delhi, India

2016-2017 **Vice-President** of Energy Society, CES, IIT Delhi, India

References

Prof. Hung D. Nguyen, *Assistant Professor*, School of Electrical and Electronic Engineering, NTU Singapore, *hunghtd@ntu.edu.sg*.

Prof. Ashu Verma, *Associate Professor*, Centre for Energy Studies, IIT Delhi, India, *averma@ces.iitd.ac.in*.

Prof. S. R. Kapoor, *Professor*, Department of Electrical Engineering, RTU Kota, India, *srkapoor@rtu.ac.in*.